

Final Project Proposal — Group 9

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1 Motivation & Overview

Over the semester, we have heavily studied the algorithmic effects on user consumption in the context of news media and political polarization. However, the dynamics of “cultural” recommendation systems present a distinct set of challenges. Recent literature suggests a growing tension between algorithmic engagement maximization and content diversity. Furthermore, a shift towards *serendipity*—the concept that users actively desire the psychological experience of exploration—has come to light.

This project proposes a comprehensive review of relevant literature (see Appendix Section 4 for our starting point) to synthesize how diversity is handled in modern recommendation systems for cultural, non-news content. After studying both the measurement of diversity and user demand for serendipity, we will investigate how these systems incorporate such exploration while maintaining user engagement.

2 Key Concepts

As part of our review, we will trace how definitions of a successful recommendation, as well as system mechanics (the recommendation engines themselves), have evolved in recent literature. Specifically:

- **System Mechanics & Key Signals:** We will begin with key definitions of implicit feedback (e.g., song completion rates, skips), the explore/exploit tradeoff, latent spaces generated via matrix factorization, and more modern methods like deep and reinforcement learning.
- **Algorithmic Metrics:** We will explore how *accuracy*, the historical baseline of recommendation engines, differs from *utility*, which captures actual user satisfaction. As part of this, we will study how diversity is measured and how short-term and long-term diversity differ.
- **User Experience:** We will define and contrast *novelty* and *serendipity*: the former of which is how new a recommendation actually is, whereas the latter refers to feelings of unexpected discovery, combining novelty with other psychological factors. A keen focus will be on how serendipity is measured.

3 Research Question

Ultimately, by synthesizing novel findings from recent literature, we will investigate how modern recommendation systems for cultural content maximally optimize user engagement while preserving diversity and the psychological demand for serendipity. Said conversely, how can algorithms prevent a collapse to algorithmic monoculture while still maintaining user engagement?

4 Appendix

4.1 Primary References

Anderson, Ashton, et al. “Algorithmic Effects on the Diversity of Consumption on Spotify.” *Proceedings of The Web Conference 2020*, 2020, pp. 2155–2165.

Johnson, Christopher C. “Logistic Matrix Factorization for Implicit Feedback Data.” *Advances in Neural Information Processing Systems (NeurIPS Workshop)*, vol. 27, 2014.

Mok, Lillio, et al. “The Dynamics of Exploration on Spotify.” *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 16, 2022, pp. 675–686.

Peng, Kenny, et al. “Reconciling the Accuracy-Diversity Trade-off in Recommendations.” *Proceedings of the ACM on Web Conference 2024*, 2024.

4.2 Additional References for Consideration

Maccatrozzo, Valentina, et al. “The Role of Serendipity in User-Curated Music Playlists.” *User Modeling and User-Adapted Interaction*, 2023.

Shakespeare, Dougal, et al. “Reframing the Filter Bubble Through Diverse Scale Effects in Online Music Consumption.” [*Journal/Conference Name TBD*], 2025.

Zhang, Yuancao, et al. “Auralist: Introducing Serendipity into Music Recommendation.” *Proceedings of the Fifth ACM International Conference on Web Search and Data Mining*, 2012, pp. 93–102.